Girls Build Bots — Junior Cybersecurity Investigator

Badge-aligned workshop guide for robotics-team-led outreach

Level: Juniors (Grades 4–5) · **Badge:** Cybersecurity — Cybersecurity Investigator **Series:** Cybersecurity (Basics · Safeguards · Investigator) **Format:** One 90–120 minute session · **Ratio:** 1 mentor per 4–5 Scouts **Who runs it:** A FIRST robotics team (FLL / FTC / FRC) leading the session. A physical robot is a nice bonus but **not required** — every activity runs on printed clue materials the team prepares in advance.

> **Troop Leaders:** This curriculum is designed to *align with* the Junior Cybersecurity badge series. Official badge requirements are in your Girl Scout Volunteer Toolkit (VTK). Verify completed activities against the official steps before awarding badges. Badges are purchased through your council shop.

The through-line: being safe and smart online

The series builds real digital literacy: understand how information travels (Basics), protect it (Safeguards), and **investigate critically** (Investigator). Juniors can follow evidence trails and reason from clues — real investigative thinking.

Vocabulary: evidence, clue, trail, verify, source, timeline, conclusion.

Cybersecurity Investigator (90–120 minutes)

Badge theme: How cyber cases get cracked — following digital trails, evaluating evidence, and reaching conclusions you can defend.

Materials

- A staged case file per team (mentor-prepared, entirely fictional): a printed suspicious message, a “sender profile” card, a hand-drawn “link preview,” and a simple event timeline
- Evidence log sheets: Clue / What it tells us / Trustworthy?
- “Verify it” station cards (check the sender, check the link spelling, check with the person it claims to be from)
- Detective props if you like — the framing works

Activity 1 — How Investigators Think (15 min)

Investigators don’t guess — they collect **evidence**, check whether each **source** is trustworthy, and build a **timeline** before reaching a conclusion. Quick warm-up: show one suspicious message; as a group, list every observable clue before deciding anything. Observation first, verdict last.

Activity 2 — The Case of the Prize Message (45 min)

The main event, in teams of 3–4. Each team receives the fictional case file: a classmate (invented) got a message saying she won a tablet — is it real? Teams work the evidence: 1. **Log each clue** on the evidence sheet (misspelled sender, urgent tone, link text doesn't match the label, timeline shows the "company" account was created yesterday) 2. **Verify** using station cards — what would checking the real company's site show? What would asking the sender directly reveal? 3. **Build the timeline** of what happened when 4. **Reach a conclusion** with the evidence that supports it

Mentors play witnesses/resources when teams "verify," feeding answers from a prepared key.

Activity 3 — Present the Case (20 min)

Each team presents like investigators: the evidence, the verification steps, the conclusion, and their recommendation ("don't click; report it; here's why"). Other teams may cross-examine with one question. The bar: **could you defend your conclusion with evidence alone?**

Activity 4 — Investigator's Habits (10 min) + Wrap (5 min)

Distill the method into three habits Scouts keep: look before trusting, verify with a second source, and conclusions need evidence. Closing: each Scout names the clue from today she'll spot fastest next time.

> **Robot moment (optional):** Share for 60 seconds how the team investigates a robot failure the same way — logging symptoms, checking one suspect at a time, and only "fixing" what the evidence points to.

Completion Checklist — Cybersecurity Investigator

By the end of the session, to earn the badge, each Scout can: - [ ] Explain how investigators use evidence, sources, and timelines - [ ] Log clues from a suspicious message and rate their trustworthiness - [ ] Verify a claim using a second source - [ ] Present a conclusion supported by evidence

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Mentor Guide (for the robotics team)

- **Prepare the case file well.** The session lives or dies on the props — 30 minutes of prep, entirely fictional, no real brands or people.
- **You don't need a robot for this.** The "robot moment" is a 60-second add-on.
- **Don't confirm early guesses.** When a team jumps to a verdict, redirect: "what evidence supports that?"
- **Play the verification stations straight.** Give factual answers from your key; let the teams do the reasoning.
- **Cross-examination stays friendly.** One respectful question per team keeps it rigorous without turning adversarial.

📣 Share Your Impact — required for teams and troops using this curriculum

If you use Girls Build Bots materials to run a workshop, please complete all three:

1. **Credit us.** Tell your Scouts and audience that the curriculum comes from Girls Build Bots, and credit "Girls Build Bots" anywhere you share about the event.
2. **Tag us.** Post about your workshop and tag **@girlsbuildbots** on Instagram.
3. **Report your results.** Email **girlsbuildbots@gmail.com** with: the number of girls who participated, the number of badges earned, and a sentence or two of feedback on how it went.

This is how the program grows and how we measure our impact — your report becomes part of the Girls Build Bots story and helps the next troop and team.

Created for the "Girls Build Bots" Girl Scout Gold Award project. Original curriculum aligned to the Junior Cybersecurity badge series; official requirements © Girl Scouts of the USA, available to troop leaders in the Volunteer Toolkit.